

The utopia style

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1. Introduction

Sometimes, your regular habits just aren't as fun anymore. Sometimes you need to trade pepperoni for a white pizza, and sometimes you need to just use something with some pizzazz. Sometimes, you need to pick a different word to start your sentences—that happens very frequently. The utopia style is designed just for that—when you need something a bit different.

The primary font in this is Erewhon, which is just a slightly extended version of Utopia, one of Adobe's original fonts. I *love* this font—I think it looks brilliant. That said, Erewhon is a highly situational font. I can't use this for a lot of everyday stuff, since it's not quite as neutral as STIX, nor does it have that “math pedigree” from Computer Modern—Erewhon is stuck playing second fiddle to these two main workhorses. That said, I think Erewhon needs to be given a fair shake; even if it's not my everyday pick, it's still a great bit of typography, and I *want* to use it more in my typesetting.

Erewhon is what Robert Bringhurst might call a “lyrical modernist” font:¹ the italic is *partially* liberated from the roman font, and it has a sort of humanist feel to it. I have to say, Erewhon is just *brimming* with craftsmanship; you can see it in the ligatures and kerning, which are far more complete than Computer Modern or STIX. There's a lot to love here. This is the style I would use when I want to subtly one-up what I'd already done—Erewhon is a statement piece which appears effortless. It's not trying to impress you, it simply *is* impressive. All of this is coupled with Merriweather sans which, while not as robust as Inter, complements Erewhon far better than I had initially expected.

2. Standard demo

The following text is taken from Lieb, E., Schultz, T., Mattis, D. (2004).² Please, go read the actual paper, it's great.

¹Bringhurst, Robert. *The Elements of Typographic Style*. 3rd ed. Point Roberts, WA: Hartley & Marks, Publishers, 2004, 15.

²Two Soluble Models of an Antiferromagnetic Chain. In: Nachtergaele, B., Solovej, J.P., Yngvason, J. (eds) *Condensed Matter Physics and Exactly Soluble Models*. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-06390-3_35

2.1. Formulation

The first model consists of N spin $1/2$'s (N even) arranged in a row and having only the nearest neighbor interactions. It is

$$H_\gamma = \sum_i [(1 + \gamma) S_i^x S_{i+1}^x + (1 - \gamma) S_i^y S_{i+1}^y],$$

where γ is a parameter characterizing the usual degree of anisotropy in the xy -plane. Because the Hamiltonian only involves the x - and y - components of the spin operators, we call this model the XY model.

To solve the XY model, we first introduce the raising and lowering operators

$$a_i^\dagger = S_i^x + i S_i^y \text{ and } a_i = S_i^x - i S_i^y$$

in terms of which the Pauli spin operators are

$$S_i^x = (a_i^\dagger + a_i)/2; S_i^y = (a_i^\dagger - a_i)/2i; S_i^z = a_i^\dagger a_i - \frac{1}{2}$$

and the Hamiltonian is

$$H_\gamma = \frac{1}{2} \sum_i [(a_i^\dagger a_{i+1} + \gamma a_i^\dagger a_{i+1}^\dagger) + \text{h.c.}].$$

The ground state Ψ_0 is the state with no elementary excitations:

$$\eta_k \Psi_0 = 0, \text{ all } k.$$

The ground-state energy, according to (A-12) is

$$E_0 = -\frac{1}{2} \sum_k \Lambda_k.$$

In the limit $N \rightarrow \infty$, the sum can be replaced by an integral giving

$$\begin{aligned} \frac{E_0}{N} &= -\frac{1}{4\pi} \int_{-\pi}^{+\pi} [1 - (1 - \gamma^2) \sin^2 k]^{1/2} dk \\ &= -\frac{1}{\pi} \mathcal{E}(1 - \gamma^2), \end{aligned}$$

where $\mathcal{E}(k^2)$ is one of the complete elliptic integrals (13).